MAT 586
Lab 8

Nolan R. H. Gagnon

1

The continuous-time logistic differential equation is

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{K}\right).$$

If the initial condition is $y(0) = y_0$, then the solution to the differential equation is

$$y(t) = \frac{Ky_0e^{rt}}{K + y_0\left(e^{rt} - 1\right)}.$$

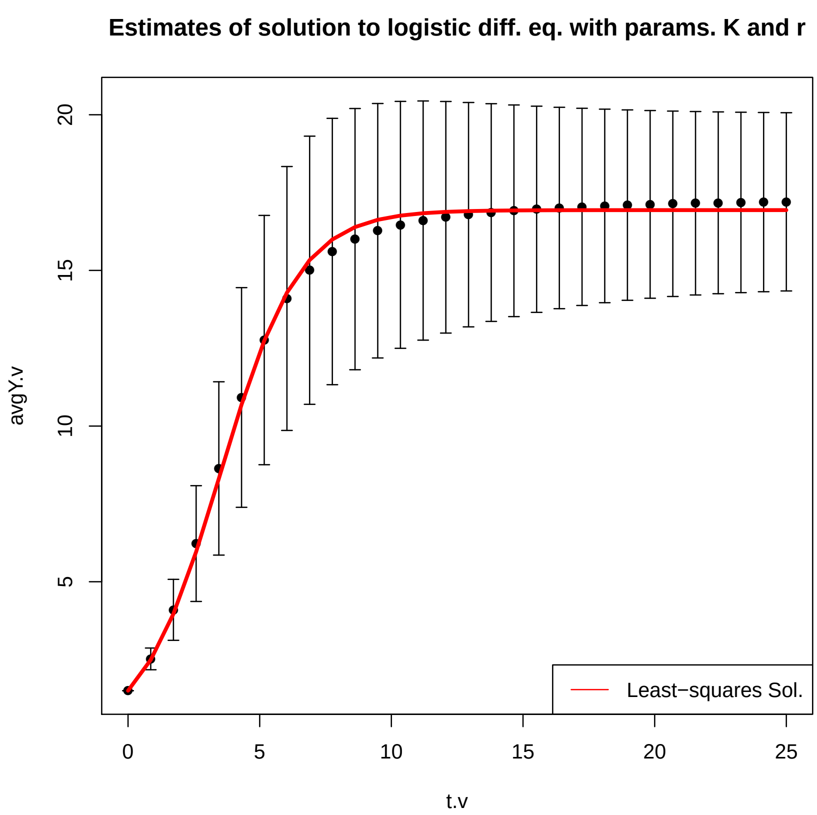

Figure 1: Estimates of solution to logistic differential equation

Figure 1 shows "average" and least-squares solutions to the differential equation with unknown K and r randomly chosen from unknown probability distributions. The least-squares estimates of K and r were $\hat{K} = 16.9370488$ and $\hat{r} = 0.6654223$. These estimates (and the figure above) were generated with the following R code.

lab8_solution.R

```r
source("log_sol.R") ## Contains function with params to estimate with least-squares optimization.
source("lab8data.R") ## Average data
library('Hmisc') ## Gives access to errbar function

par.v <- optim(c(1, 1), log_sol, t.v = t.v, y.v = avgY.v)$par

## Since y(0) = y0, it must be the case
## that y0 = avgY.v[1] = (20*y0) / 20
y0 <- avgY.v[1]

## Least-squares estimate of K
K_hat <- par.v[1]

## Least-squares estimate of r
r_hat <- par.v[2]

## Calculate the least-squares solution for y(t)
y.v <- (K_hat * y0 * exp(r_hat * t.v)) / (K_hat + y0 * (exp(r_hat * t.v) - 1))

## Plot the average solution to the logistic diff. eq.
## along with the solution found by least-squares
## optimization.
errbar(t.v, avgY.v, avgY.v + sdY.v, avgY.v - sdY.v)
lines(t.v, y.v, col = "red", lwd = 3)
title(main = "Estimates of solution to logistic diff. eq. with params. K and r")
legend("bottomright", legend = "Least-squares Sol.", lty = "solid", col = "red")
```

log_sol.R

```r
log_sol <- function(beta.v, t.v, y.v)
{
  ## Get parameters from beta vector
  K <- beta.v[1]
  r <- beta.v[2]

  ## Get y0 from average y vector
  y0 <- y.v[1]
```

```r
  ## Compute predicted values
  yhat.v <- (K * y0 * exp(r * t.v)) / (K + y0 * (exp(r * t.v) - 1))

  ## Compute residuals
  res.v <- y.v - yhat.v

  ## Return sum of squared residuals
  return(sum(res.v ^ 2))
}
```

■